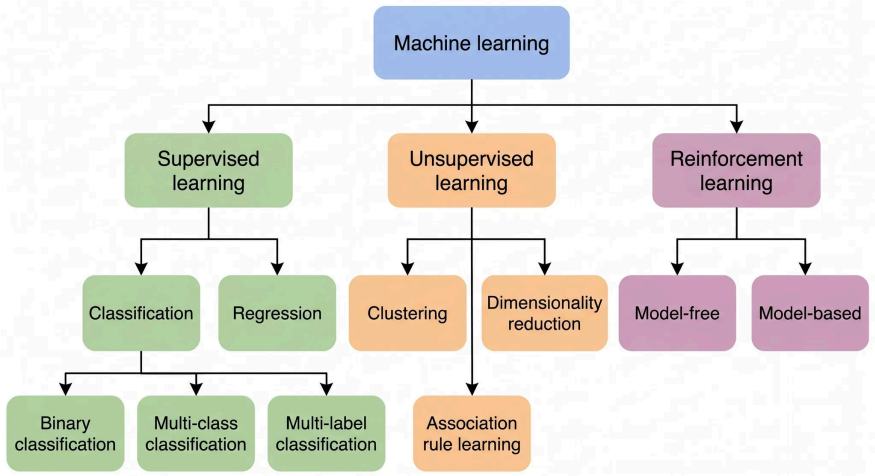
# 5. Machine Learning

Machine learning (ML) is a branch of artificial intelligence where computers learn to spot patterns and make decisions from data, rather than following rigid, hand-written rules.




## 1. The Three Main Types of Machine Learning

| Type | How It Learns | Input Data | Everyday |
|---|---|---|---|
| **Supervised Learning (SL)** | Learns with a "teacher" by studying questions paired with correct answers. | **Labeled data** (inputs + target answers) | Studying questions answers |
| **Unsupervised Learning (USL)** | Discovers hidden structures or natural groupings on its own. | **Unlabeled data** (inputs only, no answers) | Sorting a shape an their valu |
| **Reinforcement Learning (RL)** | Learns through trial and error by taking actions in an environment. | **Feedback signals** (rewards and penalties) | Training good beh them for |

- **Supervised Learning (SL):** You give the computer thousands of examples that include both the input and the right answer. Over time, the model learns the relationship between them so it can predict the answer for brand-new data it has never seen before.
- **Unsupervised Learning (USL):** The computer receives data with zero guide or answer keys. Its job is to explore the data, find underlying similarities, and group matching pieces together.
- **Reinforcement Learning (RL):** An autonomous system (called an *agent*) makes decisions inside an environment. When it takes a good action, it gets a reward; when it makes a mistake, it receives a penalty. It constantly updates its strategy to collect the highest possible reward over time.

## 2. Three Types of Supervised Learning

Supervised learning splits into tasks based on the kind of answer you want to predict:

- **Classification (Predicting Categories):** The target answer is a label, bucket, or category.
    - *How it works:* The model draws decision boundaries between distinct classes. If there are only two choices (like Yes/No), it is *binary classification*; if there are three or more, it is *multiclass classification*.
    - *Examples:* Filtering email into "Spam" vs. "Inbox", or scanning an X-ray to detect whether a bone fracture is present.
- **Regression (Predicting Numbers):** The target answer is a continuous numerical value.
    - *How it works:* The algorithm fits a line or mathematical curve through past data points to estimate quantities on an unbroken scale.
    - *Examples:* Estimating a house's selling price based on square footage and location, or predicting tomorrow's temperature in degrees.
- **Time Series Predictions (Forecasting Sequences Over Time):** The target is an ordered sequence where chronological order is the primary predictor.
    - *How it works:* Unlike standard regression where data points are treated as independent events, time series models explicitly track time-based traits like seasonality (weekly or monthly cycles), long-term trends, and lag (how yesterday's number directly impacts today's).
    - *Examples:* Forecasting electricity grid demand for the upcoming weekend, or projecting monthly inventory supply needs for retail stores.

## 3. Core Types of Unsupervised Learning (Not necessary)

Because unsupervised algorithms work without a teacher, they focus on finding patterns and relationships:

- **Clustering:** Partitioning a dataset into natural groups where items in the same group are more similar to each other than to items in other groups.
    - *How it works:* The algorithm calculates the mathematical similarity (or distance) between every data point and groups nearby neighbors together.
    - *Example:* Customer segmentation — an online store grouping shoppers into "bargain hunters", "weekend casuals", and "luxury buyers" based on their browsing behavior, without anyone manually defining those groups beforehand.
- **Association Rule Mining:** Uncovering interesting relationships and "if-then" connections between items that frequently occur together.
    - *How it works:* It scans transactions to measure how often items co-occur and checks if the presence of item $A$ reliably points to the purchase or presence of item $B$.
    - *Example:* Market Basket Analysis — discovering that supermarket shoppers who purchase hamburger patties and buns also buy ketchup 80% of the time, helping stores optimize shelf layouts and promotional bundles.

## 4. Dimensions of Data and High-Dimensional Spaces

In machine learning, **dimension** simply means the number of attributes, variables, or features recorded for each individual item.

$$\text{Attribute} = \text{Feature} = \text{Dimension} = \text{Column}$$

- **1 Dimension (1D):** Only 1 feature measured (e.g., just height). The data sits on a single line.
- **2 Dimensions (2D):** 2 features measured (e.g., height and weight). The data plots on a flat surface or chart.
- **3 Dimensions (3D):** 3 features measured (e.g., height, weight, and shoe size). The data occupies a 3D physical space (a cube).
- **High-Dimensional Space:** When you track tens, hundreds, or thousands of features per item. For instance, a medical record with 200 lab metrics is a point in 200-dimensional space, and an image made of 1,000 pixels is a point in 1,000-dimensional space.

## The Challenge: The "Curse of Dimensionality"

When data moves into high dimensions, intuitive geometry breaks down, creating three practical problems:

1. **Extreme Sparsity (Empty Space):** The volume of the space grows exponentially with each added dimension. Because the volume becomes enormous, data points end up isolated far away from one another. A dataset that comfortably fills a 2D square becomes virtually empty in a 50-dimensional space.
2. **Distance Measurements Lose Meaning:** Most ML algorithms measure how close or far two points are (using distance formulas). In very high dimensions, almost every point ends up roughly the exact same distance from every other point. When all distances look identical, clustering and classification algorithms struggle to identify which items are truly related.
3. **Overfitting and High Compute Costs:** With hundreds of variables, models require massive computational power and can easily pick up random noise as if it were a genuine pattern.

To solve this, machine learning uses **dimensionality reduction** techniques (such as Principal Component Analysis or Autoencoders). These tools compress the data into fewer dimensions while preserving the essential patterns that actually drive predictions.